

Aziz Moualhi

📍 Manouba, Tunis, Tunisia 📞 +216 29 742 817 ✉ aziz.moualhi@insat.ucar.tn
✉ azizmouelhi26@gmail.com 🌐 github.com/MoualhiAziz 🌐 Portfolio & Projects

Education

National Institute of Applied Sciences and Technology (INSAT)

Engineering Cycle — Instrumentation and Industrial Maintenance

2022 – 2027
Tunis, Tunisia

- **Key Courses:** Signal Processing, Reliability, Microcontrollers & DSP, Industrial Networks, Predictive Maintenance, Automation, Machine Learning, Intelligent Sensors, Databases, Mechanical Engineering, Heat Systems, Thermodynamics, Project Management, AI Development...

Experience

Engineering Intern — 3D Development

LRT Innovation Labs

2026 (5 months)

- Reverse-engineered 250+ automotive exhaust components via 3D scanning, rebuilding parametric CAD models in Fusion 360
- Built a **Fusion 360 add-in** (Python + Fusion 360 API) automating pipe bend-parameter extraction — cutting per-part processing from 10 minutes to 5 seconds (120x)
- Migrated parts inventory and PLM workflows from Excel into Odoo, improving traceability across engineering and manufacturing teams

Engineering Intern — Manufacturing Department

Coats Industrial Tunisia

Summer 2025 (1 month)

- Ran time studies across 4 production lines and identified bottlenecks accounting for 37% of cycle time
- Proposed lean process changes with an estimated 10–15% lead-time reduction

Projects

Full write-ups, code repositories, and live demos for every project below are on my [portfolio](#).

SolarGuard AI — Solar Panel Inspection System [Final Year Project]

2026

- 3-model AI pipeline (YOLOv11 + YOLOv9) for automated solar panel defect detection via multimodal sensor fusion of RGB/thermal drone imagery — 87.7% mAP@50; deployed as a Gradio web app. **Tools:** Python, PyTorch, OpenCV, Gradio

Industrial Defect Inspection — CNN Autoencoder with Explainable Heatmaps

2025

- Unsupervised anomaly detection (non-destructive evaluation) via reconstruction error, with z-score calibrated PASS/REVIEW/FAIL decisions and heatmap-based defect localization; deployed as a Flask web app. **Tools:** TensorFlow/Keras, OpenCV, Flask

AI for Predictive Maintenance of Turbofan Engines

2025

- LSTM-based Remaining Useful Life (RUL) prediction on NASA's C-MAPSS dataset. **Tools:** TensorFlow, Pandas, Scikit-learn

Wind Turbine Design

2024

- Designed and fabricated a 3D-printable wind turbine prototype. **Tools:** SolidWorks, 3D printing

Skills

Programming & AI: Python (TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib), Deep Learning (LSTM, CNN, Autoencoders), Image & Signal Processing, Ladder Logic, SQL, C

Industrial Systems: PLCs, Microcontrollers, SCADA basics, Embedded Systems, Sensor Fusion

Software & CAD: SolidWorks (CSWA Certified), Fusion 360, MATLAB, CAD/CAE, Git, SQL Databases, Odoo PLM, Fusion 360 API

Languages: English, French, Arabic

Areas of Expertise

Predictive Maintenance • Non-Destructive Testing (NDT) • Computer Vision • Deep Learning • Industry 4.0 • Sensor Fusion • Smart Sensors • ESP32 • Reverse Engineering • 3D Scanning & Prototyping • CAD Automation • ERP odoo • Embeded Systems